Allometric Grip Strength Norms for American Children

Morgan H. Kocher,[1] Yukiya Oba,[1] Iris F. Kimura,[1,2] Christopher D. Stickley,[1,3] Charles F. Morgan,[1] and Ronald K. Hetzler[1]

[1]Kinesiology and Rehabilitation Science Department, College of Education, University of Hawaii at Mānoa, Honolulu, Hawaii; [2]Department of Tropical Medicine, John A. Burns School of Medicine, University of Hawaii, Honolulu, Hawaii; and [3]Anatomy, Biology, and Physiology Department, John A. Burns School of Medicine, University of Hawaii, Honolulu, Hawaii

Abstract

Kocher, MH, Oba, Y, Kimura, IF, Stickley, CD, Morgan, CF, and Hetzler, RK. Allometric grip strength norms for American children. *J Strength Cond Res* 33(8): 2251–2261, 2019—To develop normative data from a large cohort of American school children (ages 6–18) for unscaled and allometrically scaled handgrip strength data that are uninfluenced by body size (body mass [BM] and stature [Ht]). Data (age, handgrip strength, BM, and Ht) were collected from the 2011–2012 and 2013–2014 National Health and Nutrition Examination Survey databases, resulting in 4,665 cases (2,384 boys and 2,281 girls). Multiple log-linear regressions were used to determine allometric exponents for BM and Ht separately for each age and sex to satisfy the common exponent and group difference principles described by Vanderburgh. Appropriateness of the allometric model was assessed through regression diagnostics, including normality and homoscedasticity of residuals. Allometrically scaled, ratio-scaled, and unscaled grip strength were then correlated with BM and Ht to examine the effectiveness of the procedure in controlling for body size. The data did not allow for development of a common exponent across age and sex that did not violate the common exponent and group difference principles. Correlations between allometrically scaled handgrip strength with BM and Ht were not significant ($p \leq 0.479$) and approached zero, unlike correlations of unscaled handgrip strength with BM and Ht ($p < 0.001$ for all), indicating that allometric scaling was successful in removing the influence of body size. Allometric scaling handgrip strength by age and sex effectively controls for body size (Ht and BM) and perhaps maturation (Ht). The allometric exponents and normative values developed can be used to compare handgrip strength within age and sex while controlling for body size.

Key Words handgrip, scaling, body size, NHANES

Address correspondence to Dr. Ronald K. Hetzler, hetzler@hawaii.edu.

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Introduction

Handgrip strength is often used as a field measurement to quickly identify individuals with strength deficits or for those presenting or at risk of various health problems. A number of studies conducted on grip strength in children have reported that stature (Ht) should be taken into consideration when examining grip strength to avoid misinterpretation (7,20,22). Furthermore, Ploegmakers et al. (20) reported a significant increase in grip strength in children with each ascending year. Both body mass (BM) and Ht have been established as significant predictors of grip strength in children. Hogrel et al. reported that Ht was directly related to strength in children and that changes in Ht occur with and are often representative of physical maturation. These authors stated that because strength is related to Ht, chronological age should not be used as the single variable to evaluate grip strength (7). However, normative handgrip strength data for American children for a given Ht do not exist and accounting for Ht alone would not account for differences between individuals of differing body sizes at a given chronological age.

Further compounding the problem of interpreting normative data in pediatric populations is the fact that Vanderburgh et al. (28) have demonstrated that grip strength should be allometrically scaled by BM. Unfortunately, normative data for children have traditionally been reported by chronological age without regard to Ht or BM (4,6,12,13,15,16,19,20). These data sets may misrepresent a child's actual percentile rank within their age group. Ramírez-Vélez et al. (21) previously attempted to account for body size by normalizing handgrip by BM (ratio scaling), but this method has been reported to overscale and simply change what was a positive correlation between BM and the scaled variable into a negative correlation (26,29). Bohannon (3) recently reported high correlations of absolute grip strength with age, height, and BM. Thus, investigation of adjustments to control for both Ht and BM by age seems justified when developing grip strength normative values for children. Nevill and Holder (14) have previously allometrically scaled handgrip strength for both Ht and BM in 12-year-old children;

TABLE 1. Demographic variables by age and sex.*

	N size		Stature (m)		Body mass (kg)		BMI (kg·m^{-2})	
Age	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
6	241	217	1.21 ± 0.06	1.20 ± 0.05	24.26 ± 5.78	24.66 ± 5.43	16.57 ± 2.75	17.01 ± 2.87
7	223	189	1.27 ± 0.06	1.25 ± 0.06	29.12 ± 8.77	27.97 ± 7.54	17.83 ± 4.05	17.57 ± 3.56
8	215	190	1.32 ± 0.07	1.32 ± 0.07	32.00 ± 9.55	32.68 ± 10.49	18.08 ± 4.05	18.50 ± 4.62
9	202	200	1.37 ± 0.07	1.38 ± 0.07	35.66 ± 9.82	38.53 ± 12.03	18.80 ± 4.04	19.93 ± 4.92
10	202	180	1.44 ± 0.07	1.45 ± 0.07	41.66 ± 12.31	42.14 ± 11.78	19.88 ± 4.55	19.96 ± 4.37
11	183	231	1.50 ± 0.08	1.51 ± 0.07	48.91 ± 14.70	48.32 ± 13.94	21.31 ± 4.98	20.98 ± 5.18
12	171	156	1.57 ± 0.08	1.56 ± 0.07	52.12 ± 15.11	56.13 ± 17.13	20.96 ± 4.76	22.81 ± 5.97
13	163	157	1.64 ± 0.08	1.59 ± 0.07	60.71 ± 17.06	57.36 ± 17.24	22.34 ± 5.43	22.57 ± 5.99
14	166	160	1.69 ± 0.08	1.60 ± 0.06	66.60 ± 19.46	61.20 ± 16.79	23.06 ± 6.01	23.91 ± 5.98
15	145	150	1.73 ± 0.07	1.61 ± 0.06	71.00 ± 20.20	63.29 ± 16.64	23.66 ± 6.01	24.44 ± 5.62
16	172	181	1.73 ± 0.07	1.61 ± 0.06	74.80 ± 20.26	64.44 ± 17.51	24.92 ± 6.27	24.68 ± 6.40
17	142	124	1.74 ± 0.08	1.61 ± 0.06	74.73 ± 17.52	65.78 ± 19.14	24.58 ± 5.34	25.07 ± 6.55
18	159	146	1.75 ± 0.07	1.61 ± 0.06	79.08 ± 22.89	65.60 ± 18.50	25.81 ± 6.84	25.04 ± 6.59
Total	2,384	2,281						

*BMI = body mass index.

however, no normative data were presented. We have previously developed grip strength norms which were allometrically scaled to remove both the influence of BM and Ht on grip strength in school children of Hawaiian lineage (9).

The National Health and Nutrition Examination Survey (NHANES) is a program started by the Centers for Disease Control in the 1960s and is designed to assess the health and nutritional status of adults and children in the United States. These data collected were used to create national standards and are publicly available for use in epidemiological studies. Data from the 2 most recent NHANES data sets (2011–2012 and 2013–2014) have included grip strength data, presenting an opportunity to create American national normative values free of influence from Ht and BM.

Normative data for handgrip strength that controls for both Ht and BM are important because of the wide range of

TABLE 2. Mean handgrip strength for boys given in various units.*

	Boys		
Age	Kilograms	Newtons	Pounds
6	10.96 ± 2.43	107.52 ± 23.81	24.11 ± 5.34
7	12.91 ± 2.83	126.62 ± 27.76	28.40 ± 6.23
8	14.42 ± 3.21	141.46 ± 31.48	31.72 ± 7.06
9	16.46 ± 3.17	161.49 ± 31.13	36.22 ± 6.98
10	18.89 ± 3.81	185.28 ± 37.37	41.55 ± 8.38
11	21.74 ± 4.72	213.28 ± 46.26	47.83 ± 10.37
12	24.49 ± 5.88	240.25 ± 57.64	53.88 ± 12.93
13	30.11 ± 6.90	295.40 ± 67.70	66.25 ± 15.18
14	35.13 ± 6.87	344.59 ± 67.38	77.28 ± 15.11
15	39.26 ± 7.05	385.19 ± 69.14	86.38 ± 15.51
16	41.30 ± 7.61	405.20 ± 74.66	90.87 ± 16.74
17	43.49 ± 7.50	426.64 ± 73.59	95.68 ± 16.50
18	44.73 ± 6.49	438.84 ± 63.63	98.41 ± 14.27

*Values given as mean ± *SD*.

TABLE 3. Mean handgrip strength for girls given in various units.*

	Girls		
Age	Kilograms	Newtons	Pounds
6	10.41 ± 2.20	102.09 ± 21.63	22.90 ± 2.20
7	11.79 ± 2.36	115.68 ± 23.17	25.94 ± 2.36
8	13.65 ± 3.08	133.87 ± 30.19	30.02 ± 3.08
9	15.70 ± 3.58	154.03 ± 35.08	34.54 ± 3.58
10	18.17 ± 3.74	178.28 ± 36.67	39.98 ± 3.74
11	20.94 ± 4.52	205.47 ± 44.36	46.08 ± 4.52
12	24.15 ± 4.92	236.96 ± 48.29	53.14 ± 4.92
13	25.92 ± 5.81	254.26 ± 57.00	57.02 ± 5.81
14	27.30 ± 4.78	267.77 ± 46.87	60.05 ± 4.78
15	28.39 ± 4.75	278.47 ± 46.59	62.45 ± 4.75
16	28.48 ± 4.68	279.37 ± 45.92	62.65 ± 4.68
17	28.91 ± 5.64	283.65 ± 55.32	63.61 ± 5.64
18	29.33 ± 5.42	287.69 ± 53.17	64.52 ± 5.42

*Values given as mean ± *SD*.

Table 4. Percentile ranks for unscaled maximum grip strength by age for boys.

Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Min	5.5	7.5	6.9	8.0	10.4	11.5	11.9	15.2	20.4	22.3	19.5	29.2	27.8
5%	7.1	8.5	9.6	10.9	12.9	14.5	16.2	20.6	24.3	28.8	29.9	31.4	34.4
10%	8.0	9.7	10.3	12.5	14.1	15.8	18.0	22.2	25.9	30.4	33.1	33.6	37.4
20%	8.8	10.5	11.3	13.8	15.8	17.8	19.3	23.8	29.6	32.7	34.9	36.0	39.2
30%	9.7	11.3	12.4	15.1	16.9	18.9	20.6	26.0	31.4	35.6	36.7	39.6	41.4
40%	10.2	12.0	13.7	15.6	17.7	19.9	22.1	27.2	33.2	37.3	38.7	41.4	42.6
50%	10.8	12.6	14.4	16.5	18.6	21.2	23.8	29.2	34.3	39.3	40.8	42.8	44.0
60%	11.3	13.3	15.3	17.4	19.5	22.5	25.3	31.2	36.5	40.8	42.6	45.0	46.2
70%	12.3	14.3	16.2	18.1	20.6	24.1	27.0	32.7	38.5	43.0	44.7	46.5	48.0
80%	13.1	15.0	17.1	19.1	21.6	26.0	29.4	36.1	41.2	45.1	47.3	49.9	50.2
90%	14.1	16.7	18.3	20.4	24.8	28.2	32.2	39.3	43.7	48.5	50.6	52.9	52.8
95%	14.9	17.9	19.4	12.5	26.2	29.8	35.3	43.3	47.5	50.4	56.1	56.2	55.7
Max	19.7	25.5	25.4	27.1	29.1	35.4	45.2	53.5	57.3	58.8	65.0	69.1	68.8

developmental progress in children of a similar age. The lack thereof makes comparisons between individuals problematic at best, and in some cases misleading. Grip strength has also recently been shown to be a strong predictor for health, and low grip strength has been correlated with several morbidities (17,18,23); therefore, accurately identifying children in need of remedial strength training is important for improving outcomes later in life. Allometric scaling of handgrip strength using multiple log-linear regressions allow for individuals of a given age group to be compared with their peers free of the influence of Ht and BM. Thus, a smaller individual, who would likely be placed in a low percentile for unscaled handgrip strength, may actually be found to be relatively stronger than many of their peers once the influence of Ht and BM on grip strength has been removed (9). Allometric scaling addresses many of these problems and would allow strength coaches or physical educators to quickly screen a group of children and identify individuals who would benefit from additional training that might otherwise go unrecognized.

Therefore, the purpose of this study was to investigate the feasibility of normative data sets derived from a large cohort of American school children ages 6–18 for unscaled handgrip strength as well as allometrically scaled handgrip strength data that are uninfluenced by body size (Ht and BM). It was hypothesized that allometric scaling would create a variable set with correlations that approached zero between the scaled variable (handgrip strength) with Ht

Table 5. Percentile ranks for unscaled maximum grip strength by age for girls.

Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Min	6.2	6.9	4.0	8.0	9.4	8.3	13.0	12.1	17.1	18.4	16.9	12.3	19.5
5%	7.1	8.5	9.2	11.0	12.4	13.9	16.5	17.9	20.1	21.0	20.9	20.1	21.6
10%	7.7	9.0	10.0	11.7	14.1	15.1	17.7	18.8	21.4	22.8	22.9	22.0	22.6
20%	8.4	9.7	11.2	12.8	15.4	17.3	19.6	21.0	23.3	24.6	24.7	24.4	24.5
30%	9.1	10.4	12.0	13.4	16.2	18.3	21.4	22.1	24.6	26.0	26.4	26.2	26.1
40%	9.7	11.0	12.7	14.5	17.0	19.4	22.6	23.9	25.5	26.8	27.3	27.6	27.4
50%	10.3	11.6	13.3	15.3	17.9	20.9	23.9	25.3	26.7	27.6	28.2	28.4	28.8
60%	10.9	12.3	14.0	15.9	18.5	21.9	25.6	26.8	28.1	28.6	29.3	29.8	30.0
70%	11.4	13.0	15.1	17.1	19.6	23.2	27.0	28.4	29.9	30.1	30.2	31.5	31.7
80%	12.2	13.8	16.1	18.2	21.2	24.9	28.5	31.3	31.2	31.7	32.4	33.3	34.2
90%	13.3	14.8	17.8	20.3	22.5	26.9	30.6	34.4	34.3	35.3	34.0	35.9	36.2
95%	14.6	15.8	19.1	23.7	23.7	28.8	32.7	35.1	35.8	38.0	37.8	39.1	39.4
Max	17.4	19.4	25.5	29.6	33.7	34.9	37.2	51.3	39.7	41.3	46.4	45.1	47.4

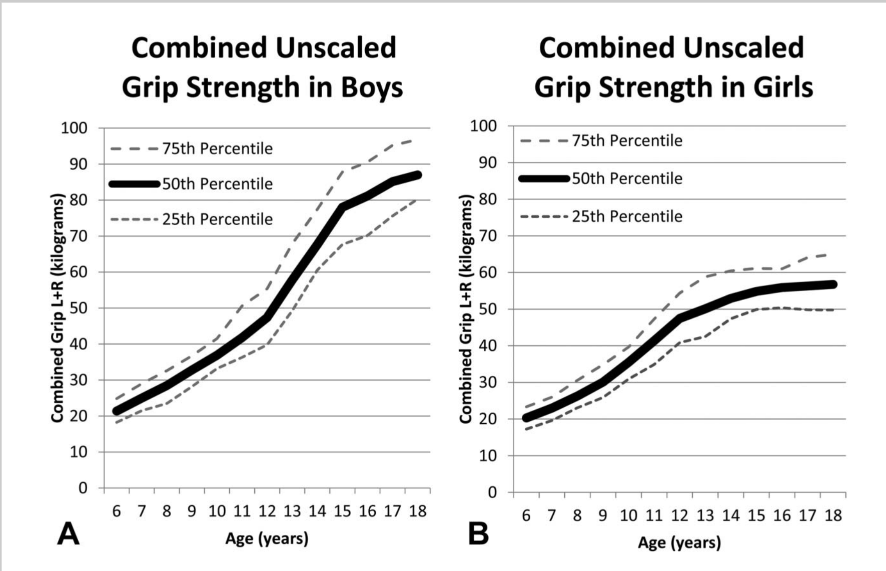

Figure 1. Combined left and right grip strength percentile normative values for boys (A) and girls (B). L = left; R = right.

and BM. It was additionally hypothesized that unscaled data or normalization using simple ratio scaling would result in significant but opposite correlations between the scaled variable with Ht and BM.

Methods

Experimental Approach to the Problem

A cross-sectional quasi-experimental design was used in this study to examine the development of unscaled and allometrically scaled normative data sets for grip strength for American school children. To control for body size, allometric exponents were determined by log-linear regression and subjected to regression diagnostics as suggested by Batterham and George (1). In addition, to eliminate potential problems with allometric scaling, the data were examined to ensure adherence to the common exponent and group difference principles, as described by Vanderburgh (25). After verifying that the conditions of the regression diagnostics were met, normative data were then allometrically scaled to control for BM and Ht. Pearson correlation coefficients were generated to verify that the allometric exponents resulted in correlations between allometrically scaled grip strength with Ht and BM that approached zero (28). Normative unscaled data and allometrically scaled data for BM and Ht by percentile ranks were then reported for American school children by sex and age.

Subjects

Participant data were collected from the NHANES online database for the years 2011–2012 and 2013–2014. Data for

Table 6. Allometric scaling exponents and percentile ranks by age for boys.*†

Age	6	7	8	9	10	11	12	13	14	15	16	17	18
B_1 (Ht)	1.511	1.487	1.752	2.171	1.733	1.660	2.770	2.268	1.853	1.221	1.438	0.847	1.106
B_2 (BM)	0.103	0.277	0.138	0.093	0.24	0.228	0.096	0.161	0.244	0.274	0.154	0.239	0.145
Min	3.06	2.11	2.94	3.87	2.46	2.62	2.01	3.34	3.27	4.51	4.79	6.47	3.32
5%	4.01	2.65	3.91	4.38	3.12	3.45	3.61	3.74	3.58	4.82	7.18	7.40	4.17
10%	4.39	2.82	4.14	4.70	3.34	3.64	3.85	3.90	3.91	5.03	7.94	7.75	4.46
20%	4.85	3.04	4.53	5.11	3.62	4.00	4.21	4.25	4.27	5.45	8.41	8.31	4.71
30%	5.15	3.17	4.88	5.43	3.76	4.18	4.39	4.52	4.38	5.69	8.80	8.81	4.95
40%	5.51	3.33	5.16	5.73	3.88	4.34	4.59	4.72	4.57	6.00	9.16	9.37	5.12
50%	5.82	3.47	5.42	6.02	4.09	4.50	4.76	4.98	4.71	6.33	9.58	9.69	5.34
60%	6.22	3.70	5.66	6.17	4.25	4.66	4.95	5.23	4.86	6.53	9.94	10.03	5.54
70%	6.56	3.92	6.00	6.43	4.44	4.88	5.16	5.49	5.03	6.71	10.28	10.50	5.75
80%	6.88	4.17	6.38	6.70	4.70	5.06	5.41	5.80	5.29	7.02	10.89	11.17	6.23
90%	7.42	4.34	7.14	7.19	4.95	5.65	5.72	6.17	5.64	7.59	11.99	11.68	6.53
95%	7.80	4.61	7.37	7.68	5.16	5.87	5.95	6.61	6.01	8.09	12.91	12.35	6.80
Max	9.32	5.18	8.32	8.02	5.79	6.80	7.06	7.52	6.77	8.89	14.89	13.82	8.01

*Ht = height; BM = body mass; Min = minimum value; Max = maximum value.
†It is inappropriate to make comparisons between age groups and sexes as the allometric exponents differ.

Table 7. Allometric scaling exponents and percentile ranks by age for girls.*†

Age	6	7	8	9	10	11	12	13	14	15	16	17	18
B_1 (Ht)	2.139	1.733	1.668	2.266	1.930	1.543	1.217	1.404	0.915	1.134	0.627	1.323	1.235
B_2 (BM)	0.107	0.214	0.214	0.185	0.219	0.278	0.327	0.373	0.340	0.284	0.318	0.203	0.263
Min	3.13	2.49	1.65	2.38	2.37	1.76	2.25	1.92	2.97	3.64	3.70	2.57	3.32
5%	3.61	3.03	2.96	2.82	2.61	2.75	2.80	2.20	3.47	4.06	4.38	4.89	4.17
10%	3.89	3.18	3.17	3.07	2.89	2.88	3.05	2.34	3.63	4.26	4.69	5.08	4.46
20%	4.22	3.40	3.48	3.34	3.04	3.25	3.29	2.51	3.87	4.46	4.96	5.64	4.71
30%	4.47	3.57	3.67	3.50	3.16	3.46	3.43	2.69	4.03	4.80	5.20	5.96	4.95
40%	4.72	3.68	3.90	3.67	3.35	3.62	3.58	2.82	4.21	4.93	5.42	6.27	5.12
50%	4.90	3.84	4.07	3.77	3.51	3.78	3.74	2.98	4.43	5.08	5.65	6.61	5.34
60%	5.18	4.04	4.26	3.91	3.64	3.93	3.92	3.13	4.62	5.29	5.82	6.83	5.54
70%	5.44	4.19	4.43	4.07	3.72	4.09	4.10	3.27	4.78	5.49	6.01	7.06	5.75
80%	5.78	4.42	4.67	4.37	3.90	4.35	4.31	3.46	4.95	5.69	6.28	7.49	6.23
90%	6.13	4.70	5.00	4.62	4.14	4.61	4.58	3.71	5.24	6.07	6.60	8.18	6.53
95%	6.61	4.88	5.36	4.96	4.35	4.90	4.75	3.82	5.35	6.23	6.95	8.50	6.80
Max	7.54	5.50	6.81	5.62	4.71	6.29	5.56	4.67	6.18	7.33	7.63	9.12	8.01

*Ht = height; BM = body mass; Min = minimum value; Max = maximum value.
†It is inappropriate to make comparisons between age groups and sexes as the allometric exponents differ.

4,894 cases between 6 and 18 years of age were used for analysis. The National Center for Health Statistics Research Ethics Review Board approved the protocol, and written informed consent was obtained from all participants and their guardians before data collection. Sample size by age group with mean and *SD*s of age, BM, Ht, and BM index (BMI) can be found in Table 1.

Procedures

National Health and Nutrition Examination Survey. The complete protocol used for collection of the NHANES data can be found in the NHANES Muscle Strength and Procedural Manual. Briefly, grip strength was assessed in both hands over 3 trials for each hand using a Takei Digital Grip Strength Dynamometer (Model T.K.K.5401) which had been adjusted to the participants' hand size. Trials were separated by 60 seconds and completed with the arm fully extended and the participant performing a fast, hard squeeze of the handle. Anthropometric data (Ht and BM) were collected using a digital stadiometer (Measurement Concepts) and a digital scale (Mettler Toledo Panther) that were routinely calibrated during data collection, as described in the NHANES Anthropometry Procedures Manual. Data on the training status of individuals were only partially available for a small subset of the data set used for this study and were not analyzed.

Allometric Scaling. The allometric scaling procedure used in this study was described by Vanderburgh et al. (27,28). Data were transformed into a log-linear model, so that linear regression could be used to solve the value of *b* (the allometric exponent) for each variable of interest. Multiple log-linear regressions were used to determine the allometric exponents for BM and Ht using the equation: $y = ax^{-b_1} * z^{-b_2}$, where a is grip strength, x is BM, and z is Ht. Dividing strength by BM^{b_1} and Ht^{b_2} yields an allometric strength index free of influence from Body Mass or Stature. The appropriateness of the allometric model was assessed through regression diagnostics, including normality and homoscedasticity of residuals as described by Batterham and George (1). The resulting allometrically scaled grip strength variables were correlated with BM and Ht to examine the effectiveness of the procedure in controlling for BM and Ht. To demonstrate that the resulting scaled variable has been corrected for the influence of body size, the correlation of the scaled variable (grip strength) with BM and Ht should approach zero (28).

Statistical Analyses

Statistical analysis was performed using SPSS (v.23, IBM, Armonk, NY, USA). Descriptive statistics and Pearson product-moment correlations were generated. Multiple log-linear regressions were performed using log grip strength as the dependent variable and log BM and log Ht as the independent variables for each sex and age group. The resulting beta weights for each independent variable were then used as the allometric exponents (b). For regression diagnostics, the Shapiro-Wilks normality test was used to evaluate the normality of the residuals; homoscedasticity was assessed using visual examination of the plot of the standardized residuals with the regression standardized predicted value by 3 experienced investigators. Outliers were identified using the least median of squares eliminator as described by Blatna and removed from analysis (2). The predicted residual sum of squares (PRESS) was also used to cross-validate the log-log regressions and to determine

Table 8. Correlations between grip strength measures and body size.*

Correlation between allometrically scaled handgrip strength with body mass and height													
Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Boys													
Body mass													
r	−0.031	0.001	−0.034	−0.029	−0.016	−0.019	0.022	0.006	−0.010	−0.059	−0.047	−0.020	−0.008
p	0.633	0.991	0.616	0.686	0.818	0.793	0.773	0.938	0.899	0.479	0.538	0.810	0.920
Height													
r	0.002	−0.007	−0.012	−0.019	−0.008	−0.003	0.017	0.007	−0.004	−0.014	−0.011	0.005	0.000
p	0.979	0.914	0.863	0.785	0.915	0.970	0.826	0.933	0.961	0.865	0.889	0.957	0.998
Girls													
Body mass													
r	−0.004	−0.013	−0.017	0.001	−0.054	−0.018	−0.021	−0.017	−0.012	0.013	−0.016	0.004	−0.018
p	0.952	0.864	0.818	0.994	0.468	0.786	0.794	0.829	0.882	0.877	0.826	0.964	0.827
Height													
r	0.002	0.002	−0.026	0.001	−0.121	−0.016	−0.020	−0.002	0.006	−0.004	0.004	−0.002	0.012
p	0.972	0.981	0.718	0.989	0.105	0.806	0.802	0.976	0.939	0.966	0.952	0.980	0.881
Correlation between ratio-scaled handgrip strength by body mass (handgrip·body mass^{-1}) with body mass and height													
Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Boys													
Body mass													
r	−0.453	−0.553	−0.577	−0.655	−0.639	−0.543	−0.591	−0.583	−0.687	−0.709	−0.703	−0.667	−0.760
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Height													
r	−0.229	−0.283	−0.283	−0.240	−0.251	−0.284	−0.131	−0.003	−0.091	−0.199	−0.117	−0.121	−0.158
p	<0.001	<0.001	<0.001	0.001	<0.001	<0.001	0.088	<0.001	0.236	0.017	0.127	0.152	0.047
Girls													
Body mass													
r	−0.500	−0.604	−0.576	−0.621	−0.613	−0.631	−0.658	−0.565	−0.704	−0.702	−0.724	−0.689	−0.697
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Height													
r	−0.158	−0.254	−0.238	−0.180	−0.146	−0.219	−0.240	−0.100	−0.158	−0.171	−0.159	−0.158	−0.145
p	0.020	<0.001	0.001	0.011	0.051	0.001	<0.001	0.211	0.045	0.037	0.032	0.080	0.081
Correlation between ratio-scaled handgrip strength by height with body mass and height													
Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Boys													
Body mass													
r	0.134	0.465	0.261	0.321	0.487	0.461	0.474	0.379	0.479	0.374	0.188	0.270	0.273
p	0.037	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.013	0.001	<0.001

Height													
r	0.187	0.394	0.298	0.390	0.453	0.440	0.580	0.446	0.426	0.243	0.162	0.085	0.142
p	0.004	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.003	0.033	0.317	0.074
Girls													
Body mass													
r	0.291	0.445	0.392	0.492	0.536	0.444	0.494	0.510	0.520	0.482	0.459	0.322	0.394
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Height													
r	0.353	0.443	0.347	0.521	0.488	0.349	0.31	0.329	0.212	0.244	0.081	0.202	0.238
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.007	0.003	0.276	0.025	0.004
Correlation between unscaled handgrip strength with body mass and height													
Age	6	7	8	9	10	11	12	13	14	15	16	17	18
Boys													
Body mass													
r	0.263	0.575	0.394	0.448	0.588	0.574	0.558	0.446	0.534	0.446	0.254	0.350	0.354
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	<0.001	<0.001
Height													
r	0.380	0.571	0.493	0.584	0.633	0.634	0.712	0.612	0.609	0.445	0.353	0.344	0.414
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Girls													
Body mass													
r	0.405	0.561	0.509	0.577	0.622	0.525	0.571	0.561	0.573	0.554	0.509	0.394	0.449
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Height													
r	0.528	0.631	0.544	0.676	0.663	0.532	0.502	0.494	0.416	0.453	0.310	0.392	0.435
p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

TABLE 9. Mean absolute change in percentile rank between unscaled and allometrically scaled normative values.

	Absolute change	
Age	Boys (%)	Girls (%)
6	9	13
7	14	17
8	12	15
9	15	16
10	16	21
11	17	16
12	19	15
13	15	14
14	17	14
15	13	14
16	10	13
17	10	10
18	11	12
Overall	14 ± 3.2	15 ± 2.7

possible changes in R^2 (PRESS R^2) and standard error of the estimate (PRESS SEE) for use in different populations using the methods described by Holiday et al. (8). Pearson correlation coefficients were generated by age and sex to verify that the allometric model–generated scaled grip strength data were not correlated with either BM or Ht (the correlations approached zero). The alpha level was set at $p \leq 0.05$.

RESULTS

Regression Diagnostics

The results of the Shapiro-Wilks test confirmed that for each age group of boys, the residuals for the log-linear regression using BM and Ht were normally distributed. Only 7-year-old girls were found not to have normally distributed residuals ($W = 0.968$, $p = 0.048$). Separate visual examination of the residual plots by 3 experienced investigators confirmed that the residuals were homoscedastic. In addition, the data were analyzed to examine for violation of the group difference principle and the common exponent principle as described by Vanderburgh (25). An attempt was made to develop common allometric exponents for all age groups and both sexes. However, the interaction between the scaled variables (BM and Ht) and sex was significantly different from zero, indicating that common exponents were not appropriate for this data set. Cross-validation of regression models was examined using the PRESS statistic. Across age groups, the PRESS R^2 revealed a mean increase of 0.022 for boys and 0.017 for girls, whereas the PRESS SEE revealed a mean shrinkage of 0.001 for both boys and girls (8).

TABLE 10. Examples of substantial and nonsubstantial scaling differences.*

				Percentile rank			
Age	Sex	Body mass (kg)	BMI (kg·m^{-2})	Unscaled (%)	Allometrically scaled (%)	Difference between ranks	
6	Male	16.7	14.80	<1	1	~1%	◂ ▸
6	Male	35.1	24.58	18	13	5%	◂ ▸
6	Female	17.5	13.63	1	7	6%	◂ ▸
6	Female	40.9	26.43	94	86	8%	◂ ▸
7	Male	18.8	14.47	33	83	50%	▲
7	Male	75.3	37.13	56	1	55%	▼
10	Male	26.2	17.63	13	87	74%	▲
10	Male	101.8	39.37	92	19	73%	▼
10	Female	27.9	16.23	20	84	64%	▲
12	Male	34.0	14.89	8	11	3%	◂ ▸
12	Male	79.8	28.07	99	99	0%	◂ ▸
12	Male	27.7	15.68	5	85	80%	▲
12	Female	129.8	44.49	85	11	74%	▼
14	Male	83.1	26.59	100	100	0%	◂ ▸
14	Male	110.2	38.68	96	92	4%	◂ ▸
14	Female	34.9	16.90	29	92	63%	▲
17	Male	59.2	18.79	65	72	8%	◂ ▸
17	Male	114.4	37.14	7	<1	7%	◂ ▸
17	Male	114.1	31.31	71	25	46%	▼
17	Female	49.8	16.74	67	62	6%	◂ ▸
18	Female	92.2	30.21	100	99	1%	◂ ▸
18	Female	106.8	36.06	75	39	36%	▼

*BMI = body mass index.

Unscaled Mean Values

To allow for easy comparison with previous or subsequent studies, mean values and *SD*s for grip strength are presented by age and sex in pounds, kilograms, and newtons in Tables 2 and 3. Percentile ranks for unscaled handgrip strength are presented in Tables 4 and 5. Unscaled maximal grip strength had a significant positive correlation with both BM (the range for boys was $r = 0.254–0.588$, $p < 0.01$ for all; the range for girls was $r = 0.394–0.622$, $p < 0.01$ for all) and Ht (the range for boys was $r = 0.344–0.712$, $p < 0.01$ for all; the range for girls was $r = 0.310–0.676$, $p < 0.01$ for all, Table 8). Unscaled mean handgrip strength increased with age (Figure 1, mean data in Tables 2 and 3, norms in Tables 4 and 5).

Ratio Scaling

Ratio-scaled handgrip strength was significantly negatively correlated with BM (the range for boys was $r = -0.453$ to -0.760, $p < 0.01$ for all; the range for girls was $r = -0.500$ to -0.724, $p < 0.01$ for all) and, to a lesser extent, Ht (the range for boys was $r = -0.003$ to -0.236, $p \leq 0.05$ for all but 12, 14, 16, and 17 years old; the range for girls was $r = -0.100$ to -0.254, $p \leq 0.05$ for all but 10, 17, and 18 years old, Table 8).

Allometric Scaling

Normative data for allometrically scaled handgrip strength are presented in Tables 6 and 7. The range of correlations between allometrically scaled handgrip strength by BM (the range for boys was $r = -0.059$ to 0.022, $p > 0.05$ for all; the range for girls was $r = -0.054$ to 0.013, $p > 0.05$ for all) and Ht (the range for boys was $r = -0.019$ to 0.017, $p > 0.05$ for all; the range for girls was $r = -0.121$ to 0.012, $p > 0.05$ for all) for each age and sex is presented in Table 8. It is important to note that comparisons between age groups and sexes are not appropriate because each age group and sex has its own specific allometric exponents.

When comparing percentile ranks of individuals between unscaled and allometrically scaled values, there was an overall mean change in percentile rank of 14 ± 3.2% for boys and 15 ± 2.7% for girls (range 9–21% when examining by age and sex, Table 9). Correlations between allometrically scaled and unscaled handgrip strength for boys ranged from 0.73 to 0.92 (mean = 0.81) and for girls, the range was 0.75–0.88 (mean = 0.78). The greatest movement in percentile ranks was seen in individuals at the extremes of BM for their group; however, not all these individuals had large shifts in their percentile ranks. Examples of selected individuals' change in percentile ranks are shown in Table 10.

Discussion

The most important finding of the current study was that allometric scaling was successful in removing the influence of both Ht and BM on handgrip strength in adolescents (as demonstrated by the correlations approaching zero between the scaled variable with Ht and BM, Table 5). Thus, allometric scaling provides an elegant solution for eliminating the effect of body size (BM and Ht) within an age group for either sex. This allows for comparison of individuals within sex and age groups with adjustments for possible differences in physical maturation (BM and Ht). Hogrel et al. (7) reported that Ht may be representative of physical maturation; because strength is related to Ht, chronological age should not be used as the single variable to evaluate grip strength. Most normative handgrip data previously presented in the literature present unscaled data which does not account for either BM or Ht (3,4,6,12,13,15,16,19), or normalized the data for BM alone through the use of ratio scaling (10,21).

Viewing the correlations presented in Table 8, it can be seen that ratio scaling alone is insufficient to remove the effect of body size and actually results in significant negative correlations similar in strength to the significant positive correlations seen between unscaled handgrip strength and body size. The allometrically scaled data presented in this study resulted in nonsignificant correlations between both BM and Ht which approached zero. These normative data can be used by both strength coaches and clinicians to identify children who should benefit from a strength training intervention. Regrettably, because of differences in the allometric exponents between groups in the current study, comparisons between sexes and age groups are inappropriate and should not be made.

Recently, Laurson et al. (10) used allometric scaling to correct for BM in children from the 2012 NHANES survey using the theoretical allometric exponent (0.67) as a common allometric exponent. The use of a common exponent on data in this study violated the group difference principle as discussed by Vanderburgh (25). Another recent article by dos Santos et al. allometrically scaled grip strength by both Ht and BM for children in Mozambique and Portugal, using their own common exponent for Ht and BM across age groups (5); however, no regression diagnostics were reported.

Data from this study were subjected to regression diagnostics as suggested by Batterham and George (1). The "litmus test" put forth by Vanderburgh (25) states that the correlations between the scaled variable and the scaling variable should approach zero and was also applied to the data. Using these procedures, it became evident that a common allometric exponent did not exist for this data set. Further calculations were then performed to develop age group– and sex-specific exponents. Fortunately, the *n* size for each group was large enough to allow for external validity (9). Subsequent examination of these exponents revealed that the sex- and age-specific exponents passed both the regression diagnostics and Vanderburgh's "litmus test," ensuring that the principles of common exponents and group differences were not violated (Table 8). In addition, cross-validation was performed using the PRESS statistic and revealed minimal shrinkage in the R^2 values and minimal inflation of the standard error of estimate.

Use of the procedures and exponents developed in this study does seem to control for body size and possibly adjust for differences in maturation. The use of individual exponents for each age group and sex precludes comparisons between groups and makes calculation of the scaled variable somewhat more tedious. Regardless, meaningful comparisons between children of the same age and sex can be made and those who would benefit from remediation can be identified.

When comparing percentile rankings between unscaled and allometrically scaled handgrip data, the mean change of an individual's percentile rank was fairly low (see bottom of Table 9). The mean absolute percentile rank change for boys ranged from 9% for 6 years old to as high as 19% for 12 years old. For girls, the mean change ranged from 10% for 17 years old to 21% for 10 years old. The mean overall change was 14% for boys and 15% for girls. Table 10 compared selected subjects with low and high BMIs, whereas some of these subjects' percentile ranks remained relatively unchanged after undergoing allometric scaling, others changed considerably. The large changes in percentile rank indicate the utility of allometric scaling in identifying individuals who are strong and those who could benefit from remediation after controlling for body size and possibly maturation (Ht).

In summary, this study used data from the NHANES 2012–2013 and 2014–2015 data sets to develop allometric exponents for handgrip strength in children. The use of common exponents for this data set was explored and found to be in violation of the principles of common exponents and group difference (25). Regression diagnostics were used to verify the normality and heteroscedasticity of the data as suggested by Batterham and George (1). In addition, when examining the correlations between the scaled variable and the scaling variables (Ht and BM), the correlations approached zero, indicating that the influence of these variables on the scaled variable had been removed as suggested by Vanderburgh et al. (28). It was concluded that the age- and sex-specific exponents were effective for controlling for body size (Ht and BM) and possibly maturation (Ht).

Practical Applications

We suggest that coaches, educators, and clinicians no longer use unscaled or ratio-scaled grip strength normative values for assessing children, rather they should use allometrically scaled data to control for body size because of the influence of both Ht and BM on grip strength percentile ranks. It has been stated that grip strength needs to be interpreted by age, sex, Ht, and BM (7,20,28). Without appropriate scaling, accounting for all these variables simultaneously would be difficult at best and perhaps impossible. The value of using allometrically scaled data is that children can be compared with strength norms that control for body size, whereas with unscaled data, the smaller children would always seem to weaker at any age. Thus, normative values allometrically scaled by Ht and BM and separated by age and sex overcome the difficulties associated with interpreting handgrip strength in children. In addition, because of the relationship established in the literature between reduced strength and increased risk of disease (11,24), these data could assist physicians and other health professionals in identifying those at increased risk of disease and steps could be taken to attenuate the risk. Comparisons of the unscaled and allometrically scaled normative data presented in Tables 4–7 can assist in quickly identifying children who would benefit from additional strength training, regardless of BM or Ht (e.g., those with allometrically scaled strength below 30th percentile). The coach, physician, or other health professional can quickly calculate the allometric strength score using a handheld scientific calculator or spreadsheet and compare it with the normative data table. It should be noted that comparisons of allometric grip strength scores between ages and sexes are not appropriate because different exponents are used for scaling. The following equation can be used to easily determine the allometric handgrip strength scores:

$$\text{Allometric grip strength} = a \Big/ \left(\left[x^{b_1} \right] * \left[z^{b_2} \right] \right),$$

where a = grip strength in kilograms (pounds/2.2), x = BM in kilograms (pounds/2.2), z = height in meters (height in inches*0.0254), b_1 = age- and sex-specific allometric exponent for BM (Tables 6 and 7), and b_2 = age- and sex-specific allometric exponent for Ht (Tables 6 and 7).

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